SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5101

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5 **Total Marks:50**

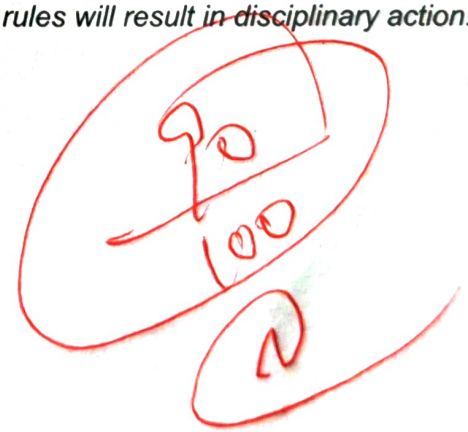
Annexure A

APPLICATION- BASED QUESTIONS

Code of Conduct:

I (Sanjana.S (192426017)) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sanjana.



Annexure A

APPLICATION- BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

   a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
   b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

   a. Deduce the most likely original message.
   b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

   a. Demonstrate how the encryption process transforms the message step-by-step.
   b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

   a. Analyze how investigators could detect the presence of hidden data within the image.
   b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEEFEAOCAIVDEN".

   a. Determine the most probable original message by reversing the encryption process.
   b. Critically evaluate why such a method is vulnerable in modern communication systems.
   c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

APPLICATION- BASED QUESTIONS.

1) Scenario: Event communication security.

A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, the message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a) Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.

This uses a Caesar cipher, where each letter is shifted by a fixed number of positions in the alphabet.

Here, the shift is 4 position forward

Alphabet :

A → B → C → D → E

B → C → D → E → F

C → D → E → F → G

Therefore, each letters is replaced by the letter 4 positions after it

original message

MEET AT NOON

Encryption

M - Q
E - I
E - I
T - X
A - E
T - X
N - R
O - S
O - S
N - R

Therefore, the encrypted message is

QIIX EX RSSR

So,

Plaintext : MEET AT NOON
shift : +4
ciphertext : QIIX EX RSSR

b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message

An authorized member who knows that the message was encrypted using a shift of 4 can decrypt it by shifting every letter 4 positions backward in the alphabet.

For example:-

Q - M
1 - E
1 - E
X - T

E - A
X - T

R - N
S - O
S - O
R - N

Therefore:

Q11X EX RSSR

becomes

MEET AT NOON :-

The decryption process reverses the encryption process. Since the authorized member knows the shift value is 4, they can easily recover the original message.

Encryption : shift letters 4 positions forward

Decryption : shift letters 4 positions backward

Thus, the original confidential event message is successfully retrieved.

2) Scenario : Intercepted suspicious message

A cybersecurity intern intercepts the coded message.

"GSRH RH HXVIG"

during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

a) Deduce the most likely original message

The most likely cipher used here is the Atbash cipher In an Atbash cipher, the alphabet is reversed :

Normal alphabet :

A B C D E F G H I J K L M N O P Q R S T U W X Y Z.

Reversed alphabet :

Z Y X W U T S R Q P O N M L K J I H G F E D C B A

using this substitution :

G → T
S → H
R → I
H → S
R → I
H → S
Z → A
H → S
X → C

coded

V - E

I - R

G - T

Therefore

GISRH RH Z HXVIG

becomes

THIS IS A SECRT

So, the most likely original message is

"THIS IS A SECRT".

b) Justify your answer by analysing repeating patterns, letters, structures, or known cipher characteristics.

The answer can be justified by examining the patterns and structure of the ciphertext.

The ciphertext is

GISRH RH Z HXVIG

It has the word pattern:

4 letters - 2letters - 1letter - 5letters

The decoded message:

THIS IS A SECRET

has exactly the same word-length pattern:

4 letters - 2letters - 1letter - 6 letters

The subsitiution also given a meaningful English sentence.

For example,

GSRH – THIS

RH – IS

Z → A

ILTXVIG → SECRET

The atbash cipher is a simple monoalphatic substitution cipher, where every letter always maps to one fixed opposite letter in the alphabet.

Some important substitutions are:

A → Z
B → Y
C → X
D → W
E → V
G → T
H → S
I → R
S → H
X → C
V → E

Applying these substitutions to the intercepted message produces a clear and meaningful sentence

Therefore, the most likely original message is:

"THIS IS A SECRET".